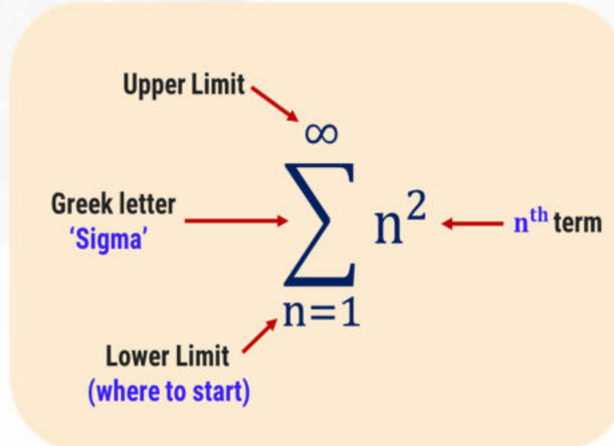
Mathematics-I
GTU#(3110014)

Unit-2

Sequence and Series



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Introduction

Series

Infinite Series

- If $u_1, u_2, \dots, u_n, \dots$ is an infinite sequence of real numbers, then the sum of terms of the sequence, $u_1 + u_2 + \cdots + u_n + \cdots$ is called an infinite series.

Notation $\sum_{n=1}^{\infty} u_n$ or $\sum u_n$

Sequence of Partial Sums

- Let $\sum u_n$ be a given infinite series.
- Consider $S_1 = u_1$

$$S_2 = u_1 + u_2$$

$$S_3 = u_1 + u_2 + u_3$$

$$\vdots$$

$$S_n = u_1 + u_2 + \cdots + u_n.$$

$$\{S_n\} = \{S_1, S_2, \ldots, S_n\}$$

$$u_1 \qquad u_1 + u_2 \qquad u_1 + u_2 + \cdots + u_n$$

Then $\{S_n\}$ defined as above is called the sequence of partial sums of the given series.

Convergence of an infinite series

- An infinite series $\sum u_n$ is said to be convergent if the sequence of partial sums $\{S_n\}$ converges to some finite number L.
- $\lim\limits_{n \to \infty} S_n = L \Rightarrow \sum_{n=1}^{\infty} u_n = L$
- If the sequence of partial sums $\{S_n\}$ diverges, then the series is said to be divergent.

Method:4 → Geometric series

Geometric Series

- An infinite series of the form $\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + ar^3 + \cdots$ is known as Geometric Series

  where, 'r' is common ratio and 'a' is the first term.

$r \in (-1,1)$

$S = \frac{a}{1-r}$

Geometric Series

Convergent if $-1 < r < 1$

Oscillating if $r \leq -1$

- $r = -1$ then **Finitely oscillating**
- $r < -1$ then **Infinitely oscillating**

Divergent if $r \geq 1$

Method:4 → Geometric Series

Example:1 Test the convergence of $\sum_{n=0}^{\infty} \frac{3^{2n}}{2^{3n}}$.

Solution: Here, $\sum_{n=0}^{\infty} \frac{3^{2n}}{2^{3n}}$

$$= \sum_{n=0}^{\infty} \frac{(3^2)^n}{(2^3)^n}$$

$$= \sum_{n=0}^{\infty} \frac{9^n}{8^n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{9}{8}\right)^n$$

$$= \left(\frac{9}{8}\right)^0 + \left(\frac{9}{8}\right)^1 + \left(\frac{9}{8}\right)^2 + \left(\frac{9}{8}\right)^3 + \cdots$$

$$= 1 + \left(\frac{9}{8}\right) + \left(\frac{9}{8}\right)^2 + \left(\frac{9}{8}\right)^3 + \cdots$$

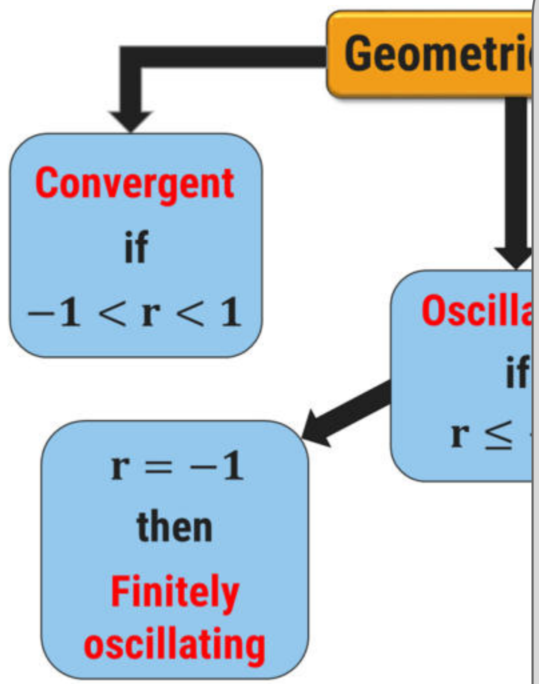
Method:4 ⇝ Example:1(Continue)

$$= 1 + \left(\frac{9}{8}\right) + \left(\frac{9}{8}\right)^2 + \left(\frac{9}{8}\right)^3 + \cdots$$ which is a geometric series,

where, $\mathbf{a} = \mathbf{1}$ and $\mathbf{r} = \dfrac{\mathbf{9/8}}{\mathbf{1}} = \dfrac{\mathbf{9}}{\mathbf{8}} > 1$

$$\therefore \mathbf{r} = \frac{\mathbf{9}}{\mathbf{8}} > 1$$

➡ Thus, the given series is divergent.



$$\mathbf{r} = \frac{\mathbf{9/8}}{\mathbf{1}} = \frac{\mathbf{9}}{\mathbf{8}}$$

$$\mathbf{r} = \frac{\left(\frac{9}{8}\right)^2}{\left(\frac{9}{8}\right)} = \frac{\mathbf{9}}{\mathbf{8}}$$

$$\mathbf{r} = \frac{\left(\frac{9}{8}\right)^3}{\left(\frac{9}{8}\right)^2} = \frac{\mathbf{9}}{\mathbf{8}}$$

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Method:4 → Geometric Series

Example:2 Define the geometric series and find the sum of $\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}}$.

Solution: Here, $\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}}$

$$= \sum_{n=1}^{\infty} \left[\frac{3^{n-1}}{6^{n-1}} - \frac{1}{6^{n-1}}\right]$$

$$= \sum_{n=1}^{\infty} \frac{3^{n-1}}{6^{n-1}} - \sum_{n=1}^{\infty} \frac{1}{6^{n-1}}$$

$$= \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n \cdots\cdots\to (1)$$

Now, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{3^{n-1}}{6^{n-1}}$

Geometric Series

- An infinite series of the form $\sum_{n=1}^{\infty} \mathbf{ar^{n-1}} = \mathbf{a + ar + ar^2 + ar^3 +} \cdots$ is known as Geometric Series where, $\mathbf{r}$ is common ratio and $\mathbf{a}$ is the first term.

Method:4 ⇝ Example:2(Continue)

➡ Now, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{3^{n-1}}{6^{n-1}}$

$$= \sum_{n=1}^{\infty} \left(\frac{3}{6}\right)^{n-1}$$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} = \left(\frac{1}{2}\right)^{0} + \left(\frac{1}{2}\right)^{1} + \left(\frac{1}{2}\right)^{2} + \cdots$$

$$= 1 + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^{2} + \cdots$$

Method:4 ⇝ Example:2(Continue)

$$= 1 + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 + \cdots$$ which is a geometric series,

where, $\mathbf{a_1} = \mathbf{1}$ and $\mathbf{r_1} = \frac{\mathbf{1/2}}{\mathbf{1}} = \frac{\mathbf{1}}{\mathbf{2}} < 1$

$$\therefore -1 < \mathbf{r_1} = \frac{\mathbf{1}}{\mathbf{2}} < 1$$

Thus, the series $\sum_{n=1}^{\infty} a_n$ is convergent and its sum is

$$S_1 = \frac{a_1}{1 - r_1} = \frac{1}{1 - \frac{1}{2}} = 2$$

Method:4 ⇝ Example:2(Continue)

$$\text{Now,} \sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{6^{n-1}}$$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{6}\right)^{n-1}$$

$$= 1 + \left(\frac{1}{6}\right) + \left(\frac{1}{6}\right)^2 + \cdots$$ which is a geometric series,

where, $\mathbf{a_2} = \mathbf{1}$ and $\mathbf{r_2} = \frac{\mathbf{1/6}}{\mathbf{1}} = \frac{\mathbf{1}}{\mathbf{6}} < 1$

$$\therefore -1 < \mathbf{r_2} = \frac{\mathbf{1}}{\mathbf{6}} < 1$$

Method:4 ⇝ Example:2(Continue)

$$\therefore -1 < \mathbf{r_2} = \frac{\mathbf{1}}{\mathbf{6}} < 1$$

- Thus, the series $\sum_{n=1}^{\infty} b_n$ is convergent and its sum is

$$S_2 = \frac{a_2}{1 - r_2} = \frac{1}{1 - \frac{1}{6}} = \frac{6}{5}$$

- Since, both the series are convergent, from equation (1)

$$\sum_{n=1}^{\infty} \frac{3^{n-1} - 1}{6^{n-1}} = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$$

Method:4 ⇝ Example:2(Continue)

$$\sum_{n=1}^{\infty} \frac{3^{n-1} - 1}{6^{n-1}} = \sum_{n=1}^{\infty} a_n - \sum_{n=1}^{\infty} b_n$$

$$\therefore S = S_1 - S_2$$

$$\therefore S = 2 - \frac{6}{5} = \frac{4}{5}$$

➡ Thus, sum of the series $\sum_{n=1}^{\infty} \frac{3^{n-1} - 1}{6^{n-1}}$ is $\frac{4}{5}$.

Method:4 Geometric Series

Example:3 Test the convergence for $5-\frac{10}{3}+\frac{20}{9}-\frac{40}{27}+\cdots$

Solution: Here, $5-\frac{10}{3}+\frac{20}{9}-\frac{40}{27}+\cdots$

$$=5\left[1-\frac{2}{3}+\frac{4}{9}-\frac{8}{27}+\cdots\right]=5\left[1-\frac{2}{3}+\left(\frac{2}{3}\right)^2-\left(\frac{2}{3}\right)^3+\cdots\right]$$

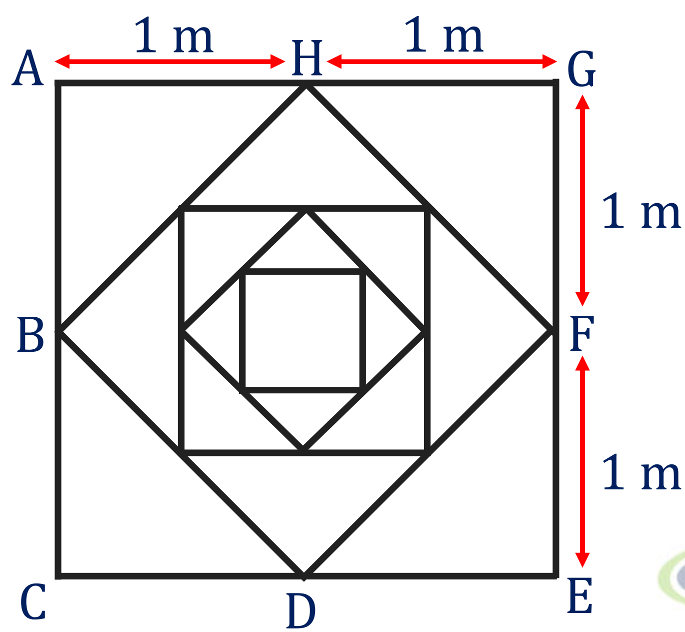
which is a geometric series, where, $\mathbf{a}=\mathbf{1}$ and $\mathbf{r}=-\frac{\mathbf{2}}{\mathbf{3}}<1$

$$\therefore -1<\mathbf{r}=-\frac{\mathbf{2}}{\mathbf{3}}<1$$

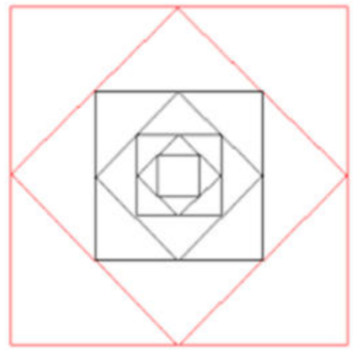
Thus, the given series is convergent.

Method:4 Geometric Series

Example:4 The figure shows the first seven of a sequence of squares. The outermost square has an area of 4m^2. Each of the other squares is obtained by joining the midpoints of the sides of the squares before it. Find the sum of areas of all squares in the infinite sequence.



Method:4 ⇝ Example:4(Continue)



Solution: Since, each square is obtained by joining the midpoints of the square before it, so the area of each square is half the area of previous square.

➡ The area of second square

$$= \frac{1}{2} \times \text{area of outermost square} = \frac{1}{2} \times 4 = 2\ m^2$$

Method:4 ⇝ Example:4(Continue)

- The area of second square

$$= \frac{1}{2} \times \text{area of outermost square} = \frac{1}{2} \times 4 = 2\ m^2$$

- The area of third square

$$= \frac{1}{2} \times \text{area of second square} = \frac{1}{2} \times 2 = 1\ m^2$$

- The area of fourth square

$$= \frac{1}{2} \times \text{area of third square} = \frac{1}{2} \times 1 = \frac{1}{2}\ m^2$$

and so on…

Method:4 $\Longrightarrow$ Example:4(Continue)

Sum of areas of all squares in the infinite sequence is

$$S = 4 + 2 + 1 + \frac{1}{2} + \cdots$$

$$\therefore S = 4\left[1 + \frac{1}{2} + \left(\frac{1}{2}\right)^2 + \cdots\right]$$ which is a geometric series,

where, $\mathbf{a} = \mathbf{1}$ and $\mathbf{r} = \frac{\mathbf{1}}{\mathbf{2}} < 1$

$$\therefore -1 < \mathbf{r} = \frac{1}{2} < 1$$

Method:4 $\Longrightarrow$ Example:4(Continue)

$$\therefore -1 < r = \frac{1}{2} < 1$$

$r \in (-1,1)$

Thus, the given series is convergent and its sum is

$$\therefore S = 4\left(\frac{a}{1-r}\right) = 4\left(\frac{1}{1-\frac{1}{2}}\right)$$

$$= 8$$

Sum of areas of all squares in the infinite sequence is $8\ m^2$.

Method:5 → Telescoping series

Telescoping Series

- A telescoping series is a series in which many terms cancel in the partial sums.

Remember this...

Partial Fraction: $a_n = \dfrac{1}{(n+1)(n+2)} = \dfrac{A}{n+1} + \dfrac{B}{n+2}$

$$\longrightarrow (n+1) = 0 \Rightarrow n = -1$$

$$\therefore A = \frac{1}{(n+2)} = \frac{1}{(-1+2)} = 1$$

$$\longrightarrow (n+2) = 0 \Rightarrow n = -2$$

$$\therefore B = \frac{1}{(n+1)} = \frac{1}{(-2+1)} = -1$$

$$\therefore a_n = \frac{1}{(n+1)(n+2)}$$

$$= \frac{1}{n+1} - \frac{1}{n+2}$$

Example:1 Check the convergence of $\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\cdots$

Solution: $$\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\cdots=\sum_{n=1}^{\infty}\frac{1}{(2n-1)\cdot(2n+1)}$$

and $$\frac{1}{(2n-1)\cdot(2n+1)}=\frac{A}{(2n-1)}+\frac{B}{(2n+1)}$$

$$=\frac{1}{2}\left[\frac{1}{(2n-1)}-\frac{1}{(2n+1)}\right]$$

$$\frac{1}{(2n-1)\cdot(2n+1)}=\frac{(2n+1)-(2n-1)}{(2n-1)\cdot(2n+1)}=\frac{2}{(2n-1)\cdot(2n+1)}$$

$$\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\cdots=\sum_{n=1}^{\infty}\frac{1}{2}\left[\frac{1}{(2n-1)}-\frac{1}{(2n+1)}\right]$$

Method:5 $\Longrightarrow$ Example:1(Continue)

$$\frac{1}{1\cdot 3}+\frac{1}{3\cdot 5}+\frac{1}{5\cdot 7}+\cdots=\sum_{n=1}^{\infty}\frac{1}{2}\left[\frac{1}{(2n-1)}-\frac{1}{(2n+1)}\right]$$

Consider, $S_m=\sum_{n=1}^{m}\frac{1}{2}\left[\frac{1}{(2n-1)}-\frac{1}{(2n+1)}\right]$

$$\therefore S_m=\frac{1}{2}\left[\begin{array}{c}\left(1-\cancel{\frac{1}{3}}\right)+\left(\cancel{\frac{1}{3}}-\cancel{\frac{1}{5}}\right)+\left(\cancel{\frac{1}{5}}-\cancel{\frac{1}{7}}\right)+\cdots+\\ \left(\cancel{\frac{1}{2m-3}}-\cancel{\frac{1}{2m-1}}\right)+\left(\cancel{\frac{1}{2m-1}}-\frac{1}{2m+1}\right)\end{array}\right]$$

Method:5 $\Longrightarrow$ Example:1(Continue)

$$\therefore S_m = \frac{1}{2}\left[\begin{array}{c}\left(1-\cancel{\frac{1}{3}}\right)+\left(\cancel{\frac{1}{3}}-\cancel{\frac{1}{5}}\right)+\left(\cancel{\frac{1}{5}}-\cancel{\frac{1}{7}}\right)+\cdots+\\ \left(\cancel{\frac{1}{2m-3}}-\cancel{\frac{1}{2m-1}}\right)+\left(\cancel{\frac{1}{2m-1}}-\frac{1}{2m+1}\right)\end{array}\right]$$

$$\therefore S_m = \frac{1}{2}\left(1-\frac{1}{2m+1}\right)$$

- Now, $\lim\limits_{m\to\infty} S_m = \frac{1}{2}\cdot\lim\limits_{m\to\infty}\left(1-\frac{1}{2m+1}\right) = \frac{1}{2}$ which is

$$\lim_{m\to\infty}\left(\frac{1}{m}\right)=0$$

- Thus, the given series is convergent and its sum is $\frac{1}{2}$.

Method:6 → Zero Test

Zero Test or Cauchy's Fundamental Test

- Suppose $\sum u_n$ is given series, where u_n is the n^{th} term of the series.

  If

  $$\lim_{n \to \infty} u_n \neq 0 \text{ or does not exist,}$$

  then the given series is **divergent**.

Method:6 Zero Test

Example:1 Check the convergence of 1). $\sum_{n=1}^{\infty} \frac{n}{2n+5}$

Solution:

$$\lim_{n \to \infty} \frac{n}{2n+5}$$

$$= \lim_{n \to \infty} \frac{\cancel{n}}{\cancel{n}\left(2 + \frac{5}{n}\right)}$$

$$= \frac{1}{2} \neq 0 \quad \left\{\because \lim_{n \to \infty} \left(\frac{1}{n}\right) = 0\right\}$$

Thus, by **Zero Test**,

the given series is divergent.

Example:1 Check the convergence of 2). $\sum_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{n}$

Solution: $\lim_{n \to \infty}\left(1+\frac{1}{n}\right)^{n} \longrightarrow (1)^{\infty}$

➔ We know that $\lim_{n \to \infty}\left(1+\frac{x}{n}\right)^{n} = e^{x}$

$$\therefore \lim_{n \to \infty}\left(1+\frac{1}{n}\right)^{n} = e \approx 2.71 \neq 0$$

➔ Thus, by **Zero Test**, the given series is divergent.

Method:6 → Zero Test

Example:3 Check the convergence of $\frac{1}{\sqrt{2}}+\frac{2}{\sqrt{5}}+\frac{4}{\sqrt{17}}+\frac{8}{\sqrt{65}}+\cdots$

Solution: $\frac{1}{\sqrt{2}}+\frac{2}{\sqrt{5}}+\frac{4}{\sqrt{17}}+\frac{8}{\sqrt{65}}+\cdots=\sum_{n=0}^{\infty}\frac{2^n}{\sqrt{4^n+1}}$

- Here, $u_n=\frac{2^n}{\sqrt{4^n+1}}$
- Now, $\lim_{n\to\infty}\frac{2^n}{\sqrt{4^n+1}}$

$$=\lim_{n\to\infty}\frac{2^n}{\sqrt{4^n}\sqrt{1+\frac{1}{4^n}}}$$

$$=\lim_{n\to\infty}\frac{2^n}{\{(2)^{\cancel{2}n}\}^{\frac{1}{\cancel{2}}}\sqrt{1+\frac{1}{4^n}}}$$

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Method:6 $\Longrightarrow$ Example:3(Continue)

$$= \lim_{n \to \infty} \frac{2^n}{\{(2)^{2\cancel{n}}\}^{\frac{1}{\cancel{2}}}\sqrt{1 + \frac{1}{4^n}}}$$

$$= \lim_{n \to \infty} \frac{\cancel{2^n}}{\cancel{2^n}\sqrt{1 + \frac{1}{4^n}}}$$

$$= 1 \neq 0 \qquad \left\{\therefore \lim_{n \to \infty}\left(\frac{1}{n}\right) = 0\right\}$$

Thus, by **Zero Test**, the given series is divergent.

Note

$$\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^{n-1} = 1 + \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 + \cdots$$

converges to $\mathbf{2}$

$$\sum_{n=1}^{\infty}\left(\frac{1}{6}\right)^{n-1} = 1 + \left(\frac{1}{6}\right) + \left(\frac{1}{6}\right)^2 + \cdots$$

converges to $\mathbf{\frac{5}{6}}$

$$\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^{n-1} - \sum_{n=1}^{\infty}\left(\frac{1}{6}\right)^{n-1} = (1-1) + \left(\frac{1}{2} - \frac{1}{6}\right) + \left\{\left(\frac{1}{2}\right)^2 - \left(\frac{1}{6}\right)^2\right\} + \cdots$$

converges to $\mathbf{2} - \mathbf{\frac{5}{6}} = \mathbf{\frac{7}{6}}$

Method:7 → Combining Series

Method:7 ➡ Combining Series

- **COMBINING SERIES:**
  - ✓ Whenever we have two convergent series, we add them term by term, subtract term by term or multiply them by constant to make new convergent series.
  - ✓ **Rules:** If $\sum a_n = a$ and $\sum b_n = b$ are convergent series then

    (1). Sum rule: $\sum(a_n + b_n) = \sum a_n + \sum b_n = a + b.$

    (2). Difference rule: $\sum(a_n - b_n) = \sum a_n - \sum b_n = a - b.$

    (3). Constant multiple rule: $\sum ka_n = k\sum a_n = k \cdot a.$
  - ✓ If $\sum a_n$ diverges, then $\sum ka_n$ also diverges.
  - ✓ If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ and $\sum(a_n - b_n)$ both diverges.
  - ✓ It is not necessary that $\sum a_n$ and $\sum b_n$ both diverges $\Rightarrow \sum(a_n + b_n)$ diverges.

Method:8 → Integral Test

Integral Test

- For the positive term series $\sum_{n=a}^{\infty} \mathbf{u_n} = \sum_{n=a}^{\infty} \mathbf{f(n)}$, if $f(n)$ is decreasing, integrable and

1) $\int_{a}^{\infty} f(x)\,dx$ is **finite**, then the given series is **convergent**.

2) $\int_{a}^{\infty} f(x)\,dx$ is **infinite**, then the given series is **divergent**.

Method:8 → Integral Test

Example:1 Check the convergence of $\sum_{n=1}^{\infty} n \cdot e^{-n^2}$.

Solution: Let, $\sum_{n=1}^{\infty} n \cdot e^{-n^2} = \sum_{n=1}^{\infty} f(n)$

→ Here, $f(n) = n \cdot e^{-n^2}; n \geq 1$

Let $f(x) = x \cdot e^{-x^2}; x \geq 1$

$\therefore f(x) = x \cdot e^{-x^2}$ is continuous and positive.

→ Now, for $x = 1, f(1) = \frac{1}{e} \approx 0.37$

Method:8 $\Longrightarrow$ Example:1(Continue)

- Now, for $x = 1, f(1) = \dfrac{1}{e} \approx 0.37$

  for $x = 2, f(2) = \dfrac{2}{e^4} \approx 0.04$

  for $x = 3, f(3) = \dfrac{3}{e^9} \approx 0.0004$ and so on.

- Thus, $f(x)$ is decreasing function.

  Therefore, the integral test is applicable.

- $$\int_{x=1}^{\infty} x \cdot e^{-x^2}\, dx = \lim_{m \to \infty} \int_{1}^{m} x \cdot e^{-x^2}\, dx$$

Method:8 $\Longrightarrow$ Example:1(Continue)

$$\int_{x=1}^{\infty} x \cdot e^{-x^2}\,dx = \lim_{m \to \infty} \int_{1}^{m} x \cdot e^{-x^2}\,dx$$

$$= \lim_{m \to \infty} \left\{ -\frac{1}{2} \left(\int_{1}^{m} (-2x) \cdot e^{-x^2}\,dx \right) \right\}$$

$$= \lim_{m \to \infty} \left\{ -\frac{1}{2} \left(e^{-x^2} \right)_{1}^{m} \right\}$$

$$= -\frac{1}{2} \left\{ \lim_{m \to \infty} \left(e^{-m^2} - e^{-1} \right) \right\}$$

$$\boxed{\int e^{f(x)} f'(x)\,dx = e^{f(x)} + c}$$

Method:8 $\Longrightarrow$ Example:1(Continue)

$$= -\frac{1}{2}\left\{\lim_{m\to\infty}\left(e^{-m^2} - e^{-1}\right)\right\}$$

$$= -\frac{1}{2}\left\{\lim_{m\to\infty}\left(\frac{1}{e^{m^2}} - \frac{1}{e^{1}}\right)\right\}$$

$$= -\frac{1}{2}\left\{\left(0 - \frac{1}{e}\right)\right\} = \frac{1}{2e}$$ which is finite.

$$\lim_{x\to\infty} e^{x} = \infty$$

Thus, by **integral test**, the given series is convergent.

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Method:8 Integral Test

Example:2 Test the convergence of $\sum_{n=1}^{\infty} \frac{2 \cdot \tan^{-1} n}{1+n^2}$.

Solution: Let, $\sum_{n=1}^{\infty} \frac{2 \cdot \tan^{-1} n}{1+n^2} = \sum_{n=1}^{\infty} f(n)$

Here, $f(n) = \frac{2 \cdot \tan^{-1} n}{1+n^2}; n \geq 1$

Let, $f(x) = \frac{2 \cdot \tan^{-1} x}{1+x^2}; x \geq 1$

$\therefore$ $f(x)$ is continuous and positive.

Method:8 $\Longrightarrow$ Example:2(Continue)

- Now, for $x = 1, f(1) = \dfrac{2 \cdot \tan^{-1}(1)}{2} \approx 0.78$

  For $x = 2, f(2) = \dfrac{2 \cdot \tan^{-1}(2)}{1 + (2)^2} \approx 0.44$

  For $x = 3, f(3) = \dfrac{2 \cdot \tan^{-1}(3)}{1 + (3)^2} \approx 0.25$ and so on.

- Thus, $f(x) = \dfrac{2 \cdot \tan^{-1}(x)}{1 + (x)^2}; x \geq 1$ is decreasing function.

  Therefore, the integral test is applicable.

Now, $\displaystyle\int_{x=1}^{\infty} f(x)\,dx = \int_{x=1}^{\infty} \frac{2\cdot\tan^{-1}(x)}{1+x^2}\,dx$

$$f(x) = \tan^{-1}(x)$$
$$f'(x) = \frac{1}{1+x^2}$$
$$n = 1$$

$$= \lim_{m\to\infty} \int_{1}^{m} \frac{2\cdot\tan^{-1}(x)}{1+x^2}\,dx$$

$$= \lim_{m\to\infty} 2\left\{\frac{[\tan^{-1}(x)]^{1+1}}{1+1}\right\}_1^m$$

$$= \lim_{m\to\infty} 2\left\{\frac{[\tan^{-1}(x)]^{2}}{2}\right\}_1^m$$

$$\int [\,f(x)\,]^n\, f'(x)\,dx = \frac{[\,f(x)\,]^{n+1}}{n+1} + c;\; n > -1$$

Method:8 $\Longrightarrow$ Example:2(Continue)

$$= \lim_{m \to \infty} 2\left\{\frac{[\tan^{-1}(x)]^2}{2}\right\}_1^m$$

$$= \lim_{m \to \infty} \{[\tan^{-1}(m)]^2 - [\tan^{-1}(1)]^2\}$$

$$= \lim_{m \to \infty} \left\{[\tan^{-1}(m)]^2 - \left[\frac{\pi}{4}\right]^2\right\}$$

$$= \lim_{m \to \infty} \{[\tan^{-1}(m)]^2\} - \frac{\pi^2}{16}$$

$$= \left(\frac{\pi}{2}\right)^2 - \frac{\pi^2}{16}$$

$$= \frac{\pi^2}{4} - \frac{\pi^2}{16}$$

$= \dfrac{3\pi^2}{16}$ which is finite.

Thus, by **integral test**,
the given series is convergent.

p-series Test

- The series $\sum_{n=a}^{\infty} \frac{1}{n^p}$ is known as p-series.

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$$

Convergent if $p > 1$

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

Divergent if $p \leq 1$

Method:9 → Direct Comparison Test

Direct Comparison Test

- If two given positive term series $\sum \mathbf{u_n}$ & $\sum \mathbf{v_n}$ with $\mathbf{0} \leq \mathbf{u_n} \leq \mathbf{v_n}, \forall\, n \in \mathbb{N}$ and if

1) $\sum \mathbf{v_n}$ is **convergent**, then $\sum \mathbf{u_n}$ is **convergent**.

2) $\sum \mathbf{u_n}$ is **divergent**, then $\sum \mathbf{v_n}$ is **divergent**.

Example:1 Test the convergence of (i). $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+1}}$.

Solution: Here, $u_n = \frac{1}{\sqrt{n^3+1}}$

Let, $v_n = \frac{1}{\sqrt{n^3}}$

$$\therefore \frac{1}{\sqrt{n^3+1}} \leq \frac{1}{\sqrt{n^3}}; \forall\, n \geq 1$$

$$\frac{1}{5} = 0.2$$

$$\frac{1}{2} = 0.5$$

$$\therefore \frac{1}{5} < \frac{1}{2}$$

$$\therefore \frac{1}{\sqrt{n^3+1}} \le \frac{1}{\sqrt{n^3}} ; \forall\, n \ge 1$$

If **bigger** series is **convergent**, then **smaller** series is **convergent**.

$$\therefore u_n \le v_n ; \forall\, n \ge 1$$

Also, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3}}$

$= \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ which is a p-series with $\mathbf{p = \frac{3}{2} > 1}$

Method:9 ⇝ Example:1(i) (Continue)

$$\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$$ which is a p-series with $\mathbf{p = \frac{3}{2} > 1}$

$\mathbf{u_n \leq v_n \,;\, \forall\, n \geq 1}$

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3}}$ is convergent.

➡ Thus, by **direct comparison test**, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+1}}$ is convergent.

Method:9 Direct Comparison Test

Example:1 Test the convergence of (ii). $\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n}-1}$.

Solution: Here, $u_n = \frac{1}{2\sqrt{n}-1}$

Let, $v_n = \frac{1}{2\sqrt{n}}$

If **smaller** series is **divergent**, then **bigger** series is **divergent**.

$$\frac{1}{2\sqrt{n}-1} \geq \frac{1}{2\sqrt{n}} \; ; \forall \; n \geq 1$$

Method:9 ⇝ Example:1(ii) (Continue)

$$\frac{1}{2\sqrt{n}-1} \geq \frac{1}{2\sqrt{n}} \; ; \forall\, n \geq 1$$

- Also, $\displaystyle\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{2\sqrt{n}}$

$$= \frac{1}{2}\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} \quad \cdots\cdots\cdots\to \quad (1)$$

- Now, $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$ is a p-series with $\mathbf{p = \frac{1}{2} < 1}$

Method:9 ⇝ Example:1(ii) (Continue)

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$ is divergent.

From equation (1), the series $\frac{1}{2}\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$ is divergent.

$(\because$ if $\sum a_n$ is diverges, then $\sum k \cdot a_n$ is also diverges.)

⇒ Now, $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$ is a p-series with $\mathbf{p = \frac{1}{2} \not< 1}$

⇒ Thus, by **direct comparison test**, $\sum_{n=1}^{\infty} \frac{1}{2\sqrt{n}-1}$ is divergent.

Method:9 → Direct Comparison Test

Example:1 Test the convergence of (iii). $\sum_{n=1}^{\infty} \frac{1}{3^n+1}$.

Solution: Here, $u_n = \frac{1}{3^n+1}$

→ Let, $v_n = \frac{1}{3^n}$

$$\therefore \frac{1}{3^n+1} \leq \frac{1}{3^n} \;;\forall\, n \geq 1$$

If **bigger** series is **convergent**,
then **smaller** series is **convergent**.

$$\therefore \frac{1}{3^n+1} \le \frac{1}{3^n} \; ; \forall\, n \ge 1$$

➡ Now, $\displaystyle\sum_{n=1}^{\infty} \frac{1}{3^n} = \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n$

$$= \left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 \cdots$$ which is a geometric series,

where, $\mathbf{a} = \frac{1}{3}$ and $\mathbf{r} = \frac{\left(\frac{1}{3}\right)^2}{\frac{1}{3}} = \frac{\mathbf{1}}{\mathbf{3}} < 1$

Method:9 ⇝ Example:1(iii) (Continue)

where, $\mathbf{a} = \frac{1}{3}$ and $\mathbf{r} = \dfrac{\left(\frac{1}{3}\right)^2}{\frac{1}{3}} = \frac{\mathbf{1}}{\mathbf{3}} < 1$

$$\therefore -1 < \mathbf{r} = \frac{\mathbf{1}}{\mathbf{3}} < 1$$

$\therefore \sum_{n=1}^{\infty} \frac{1}{3^n}$ is convergent.

➡ Thus, by **direct comparison test**, $\sum_{n=1}^{\infty} \frac{1}{3^n + 1}$ is convergent.

Remember this…

$$\sum_{n=1}^{\infty} \frac{\log n}{n^p}$$

$p \leq 1$

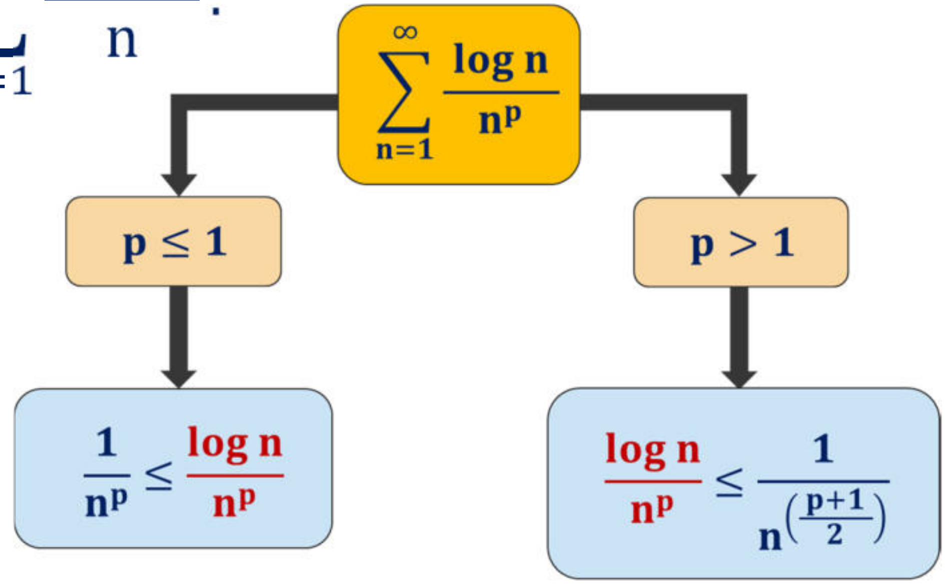
$$\frac{1}{n^p} \leq \frac{\log n}{n^p}$$

$p > 1$

$$\frac{\log n}{n^p} \leq \frac{1}{n^{\left(\frac{p+1}{2}\right)}}$$

Method:9 → Direct Comparison Test

Example:2 Check the convergence of (i). $\sum_{n=1}^{\infty} \frac{\log n}{n}$.



Solution: Here, $u_n = \frac{\log n}{n}$

→ We know that, for $p \le 1$

$$\frac{1}{n^p} \le \frac{\log n}{n^p}. \text{ Here, } p = 1$$

$$\therefore \frac{1}{n} \le \frac{\log n}{n} \;; \forall\, n \ge 1 \quad \text{where, } v_n = \frac{1}{n}$$

$$\therefore \frac{1}{n} \le \frac{\log n}{n} \; ; \forall\, n \ge 1 \quad \text{where, } v_n = \frac{1}{n}$$

➡ Also, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n}$ which is a p-series with $\mathbf{p = 1}$

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

➡ Thus, by **direct comparison test**, $\sum_{n=1}^{\infty} \frac{\log n}{n}$ is divergent.

Method:9 → Direct Comparison Test

Example:2 Check the convergence of (ii). $\sum_{n=1}^{\infty} \frac{\log n}{n^2}$.

Solution: Here, $u_n = \frac{\log n}{n^2}$

→ We know that, for $p > 1$

$$\frac{\log n}{n^p} \le \frac{1}{n^{\left(\frac{p+1}{2}\right)}}$$. Here, $p = 2$

$$\therefore \frac{\log n}{n^2} \le \frac{1}{n^{\frac{3}{2}}} ; \forall\, n \ge 1$$ where, $v_n = \frac{1}{n^{\frac{3}{2}}}$

$\sum_{n=1}^{\infty} \frac{\log n}{n^p}$

$p \le 1$: $\frac{1}{n^p} \le \frac{\log n}{n^p}$

$p > 1$: $\frac{\log n}{n^p} \le \frac{1}{n^{\left(\frac{p+1}{2}\right)}}$

$$\therefore \frac{\log n}{n^2} \le \frac{1}{n^{\frac{3}{2}}}\,; \forall\, n \ge 1, \text{where}, v_n = \frac{1}{n^{\frac{3}{2}}}$$

Also, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ which is a p-series with $\mathbf{p = \frac{3}{2} > 1}$

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ is convergent.

Thus, by **direct comparison test**, $\sum_{n=1}^{\infty} \frac{\log n}{n^2}$ is convergent.

Method:10 ➡ Limit Comparison Test

Limit Comparison Test

- If two positive term series $\sum u_n$ (given) & $\sum v_n$ (choose) be such that

$$\lim_{n \to \infty} \frac{u_n}{v_n} = L$$ and

1) If L is **finite and non-zero**, then both series $\sum u_n$ & $\sum v_n$ are **convergent** or **divergent** together.
2) If $L = 0$ and if $\sum v_n$ is **convergent**, then $\sum u_n$ is **convergent**.
3) If $L = \infty$ and if $\sum v_n$ is **divergent**, then $\sum u_n$ is **divergent**.

Example:1 Check the convergence of $\sum_{n=1}^{\infty} \frac{3n^2 + 5n}{7 + n^4}$.

Solution: Here, $u_n = \frac{3\boxed{n^2} + 5n}{7 + \boxed{n^4}}$

Let, $v_n = \frac{n^2}{n^4} = \frac{1}{n^2}$

Now, $\lim_{n \to \infty} \frac{u_n}{v_n} = \lim_{n \to \infty} \frac{(3n^2 + 5n) \cdot n^2}{7 + n^4}$

Now, $\lim_{n \to \infty} \frac{u_n}{v_n} = \lim_{n \to \infty} \frac{n^2 \cdot (3n^2 + 5n)}{7 + n^4}$

$$= \lim_{n \to \infty} \frac{\cancel{n^2} \cdot \cancel{n^2}\left(3 + \frac{5\cancel{n}}{n^{\cancel{2}}}\right)}{\cancel{n^4}\left(\frac{7}{n^4} + 1\right)}$$

$$= \lim_{n \to \infty} \frac{\left(3 + \frac{5}{n}\right)}{\left(\frac{7}{n^4} + 1\right)}$$

$$= \frac{(3 + 0)}{(0 + 1)}$$

$$= 3$$

which is finite and non-zero

$= 3$ which is finite and non-zero

Therefore, both the series are convergent or divergent together.

➡ Now, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ which is a p-series with $\mathbf{p = 2 > 1}$

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

➡ Thus, by **limit comparison test**, $\sum_{n=1}^{\infty} \frac{3n^2+5n}{7+n^4}$ is convergent.

1) If $\mathbf{L}$ is **finite and non-zero**, then both series $\sum \mathbf{u_n}$ & $\sum \mathbf{v_n}$ are **convergent** or **divergent** together.

2) If $\mathbf{L = 0}$ and if $\sum \mathbf{v_n}$ is **convergent**, then $\sum \mathbf{u_n}$ is **convergent**.

3) If $\mathbf{L = \infty}$ and if $\sum \mathbf{v_n}$ is **divergent**, then $\sum \mathbf{u_n}$ is **divergent**.

Example:2 Check the convergence of (i). $\sum_{n=1}^{\infty} \frac{1}{2^n - 3}$.

Solution: Here, $u_n = \frac{1}{2^n - 3}$

Let, $v_n = \frac{1}{2^n}$

Now, $\lim_{n \to \infty} \frac{u_n}{v_n} = \lim_{n \to \infty} \frac{\frac{1}{2^n - 3}}{\frac{1}{2^n}}$

Now, $\lim\limits_{n\to\infty}\frac{u_n}{v_n} = \lim\limits_{n\to\infty}\frac{\frac{1}{2^n-3}}{\frac{1}{2^n}}$

$$= \lim_{n\to\infty}\frac{2^n}{2^n-3}$$

$$= \lim_{n\to\infty}\frac{\cancel{2^n}}{\cancel{2^n}\left(1-\frac{3}{2^n}\right)}$$

$$= \frac{1}{(1-0)} = 1$$ which is finite and non-zero

$$= \frac{1}{\left(1-\frac{3}{2^{\infty}}\right)}$$

$$= \frac{1}{\left(1-\frac{3}{\infty}\right)}$$

$= 1$ which is finite and non-zero

Therefore, both the series are convergent or divergent together.

Now, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{2^n}$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$$

$$= \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 \dots$$

Method:10 ⤳ Example:2(i)(Continue)

$$= \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 \cdots$$ which is a geometric series,

where, $\mathbf{a} = \frac{1}{2}$ and $\mathbf{r} = \frac{\left(\frac{1}{2}\right)^2}{\frac{1}{2}} = \frac{\mathbf{1}}{\mathbf{2}} < 1$ $\qquad \therefore -1 < \mathbf{r} = \frac{\mathbf{1}}{\mathbf{2}} < 1$

$\therefore \sum_{n=1}^{\infty} \frac{1}{2^n}$ is convergent.

➡ Thus, by **limit comparison test**, $\sum_{n=1}^{\infty} \frac{1}{2^n - 3}$ is convergent.

Method:10 → Limit Comparison Test

Example:2 Check the convergence of (ii). $\sum_{n=1}^{\infty} \frac{3 + 2 \cdot \sin(n)}{n^3}$.

Solution: Here, $u_n = \frac{3 + 2 \cdot \sin(n)}{n^3}$

→ Let, $v_n = \frac{n}{n^3} = \frac{1}{n^2}$

→ Now, $\lim_{n \to \infty} \frac{u_n}{v_n} = \lim_{n \to \infty} \frac{\frac{3 + 2 \cdot \sin(n)}{n^3}}{\frac{1}{n^2}}$

Method:10 ⇝ Example:2(ii)(Continue)

Now, $$\lim_{n \to \infty} \frac{u_n}{v_n} = \lim_{n \to \infty} \frac{\frac{3 + 2 \cdot \sin(n)}{n^3}}{\frac{1}{n^2}}$$

$$= \lim_{n \to \infty} \frac{n^2 \cdot \{3 + 2 \cdot \sin(n)\}}{n^3}$$

$$= \lim_{n \to \infty} \frac{3 + 2 \cdot \sin(n)}{n}$$

$$= \lim_{n \to \infty} \frac{3}{n} + 2 \lim_{n \to \infty} \frac{\sin(n)}{n}$$

$$= 0$$

$$\left\{\because \lim_{n \to \infty} \frac{\sin(n)}{n} = 0\right\}$$

$$\& \left\{\lim_{n \to \infty} \frac{1}{n} = 0\right\}$$

$= 0$ $\quad\quad \left\{\because \lim_{n\to\infty} \frac{\sin(n)}{n} = 0\right\}$ & $\left\{\lim_{n\to\infty} \frac{1}{n} = 0\right\}$

Thus, by the **limit comparison test**, if $\sum v_n$ is convergent, then $\sum u_n$ is convergent.

➡ Now, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ which is a p-series with $\mathbf{p = 2 > 1}$

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

➡ Thus, the given series $\sum_{n=1}^{\infty} \frac{3 + 2\cdot\sin(n)}{n^3}$ is convergent.

1) If $\mathbf{L}$ is **finite and non-zero**, then both series $\sum \mathbf{u_n}$ & $\sum \mathbf{v_n}$ are **convergent** or **divergent** together.

2) If $\mathbf{L = 0}$ and if $\sum \mathbf{v_n}$ is **convergent**, then $\sum \mathbf{u_n}$ is **convergent**.

3) If $\mathbf{L = \infty}$ and if $\sum \mathbf{v_n}$ is **divergent**, then $\sum \mathbf{u_n}$ is **divergent**.

Example:2 Check the convergence of (iii). $\sum_{n=1}^{\infty}\left[10^{\frac{1}{n}}-1\right]$.

Solution: Here, $u_n = 10^{\frac{1}{n}} - 1$

Let, $v_n = \dfrac{1}{n}$

Now, $\lim\limits_{n\to\infty} \dfrac{u_n}{v_n} = \lim\limits_{n\to\infty} \dfrac{10^{\frac{1}{n}} - 1}{\frac{1}{n}}$

Method:10 ⇝ Example:2(iii)(Continue)

Now, $\lim\limits_{n \to \infty} \frac{u_n}{v_n} = \lim\limits_{n \to \infty} \frac{10^{\frac{1}{n}} - 1}{\frac{1}{n}} \longrightarrow = \frac{10^0 - 1}{0} \left(\frac{0}{0}\right)$

Applying L'Hospital's rule

$$= \lim_{n \to \infty} \frac{10^{\frac{1}{n}} \log 10 \cdot \cancel{\left(-\frac{1}{n^2}\right)}}{\cancel{\left(-\frac{1}{n^2}\right)}}$$

$$= \lim_{n \to \infty} 10^{\frac{1}{n}} \log 10$$

$$= \lim_{n \to \infty} 10^{\frac{1}{n}} \log 10$$

$$= \log 10 \text{ which is finite and non-zero} \qquad \left\{ \because \lim_{n \to \infty} \left(\frac{1}{n} \right) = 0 \right\}$$

Therefore, by **limit comparison test** both the series are convergent or divergent together.

- Now, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n}$ which is a p-series with $p = 1$

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

1) If L is **finite and non-zero**, then both series $\sum u_n$ & $\sum v_n$ are **convergent** or **divergent** together.
2) If $L = 0$ and if $\sum v_n$ is **convergent**, then $\sum u_n$ is **convergent**.
3) If $L = \infty$ and if $\sum v_n$ is **divergent**, then $\sum u_n$ is **divergent**.

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Method:10 ⇝ Example:2(iii)(Continue)

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

➡ Thus, by **limit comparison test** the given series $\sum_{n=1}^{\infty} [10^{\frac{1}{n}} - 1]$ is divergent.

Method:10 → Limit Comparison Test

Example:3 Check the convergence of (i). $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}-\sqrt{n}}{n}$.

Solution: Here, $u_n = \frac{\sqrt{n+1}-\sqrt{n}}{n}$

$$\therefore u_n = \frac{\boxed{\sqrt{n+1}-\sqrt{n}}}{n} \cdot \frac{\boxed{\sqrt{n+1}+\sqrt{n}}}{\sqrt{n+1}+\sqrt{n}}$$

$$\therefore u_n = \frac{\left(\sqrt{n+1}\right)^2 - (\sqrt{n})^2}{n\left(\sqrt{n+1}+\sqrt{n}\right)}$$

Method:10 ⇝ Example:3(i)(Continue)

$$\therefore u_n = \frac{\left(\sqrt{n+1}\right)^2 - \left(\sqrt{n}\right)^2}{n\left(\sqrt{n+1}+\sqrt{n}\right)}$$

$$\therefore u_n = \frac{n+1-n}{n\left(\sqrt{n+1}+\sqrt{n}\right)}$$

$$\therefore u_n = \frac{1}{n \cdot \sqrt{n}\left(\sqrt{1+\frac{1}{n}}+1\right)}$$

$$\therefore u_n = \frac{1}{n^{\frac{3}{2}}\left(\sqrt{1+\frac{1}{n}}+1\right)}$$

➡ Let, $v_n = \frac{1}{n^{\frac{3}{2}}}$

➡ Now, $\lim_{n \to \infty} \frac{u_n}{v_n}$

$$= \lim_{n \to \infty} \frac{\cancel{n^{\frac{3}{2}}}}{\cancel{n^{\frac{3}{2}}}\left(\sqrt{1+\frac{1}{n}}+1\right)}$$

$$= \frac{1}{\left(\sqrt{1+0}+1\right)}$$

Method:10 ⇝ Example:3(i)(Continue)

$$= \frac{1}{\left(\sqrt{1+0}+1\right)}$$

$= \frac{1}{2}$ which is finite and non-zero

Therefore, by **limit comparison test** both the series are convergent or divergent together.

- Now, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ which is a p-series with $\mathbf{p = \frac{3}{2} > 1}$

  So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ is convergent.

Method:10 ⇝ Example:3(i)(Continue)

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ is convergent.

➡ Thus, by **limit comparison test** the given series $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}-\sqrt{n}}{n}$

is convergent.

Example:3 Check the convergence of (ii). $\sum_{n=1}^{\infty} \sqrt{n^4+1} - \sqrt{n^4-1}.$

Solution: Here, $u_n = \sqrt{n^4+1} - \sqrt{n^4-1}$

$$\therefore u_n = \boxed{\sqrt{n^4+1} - \sqrt{n^4-1}} \cdot \frac{\boxed{\sqrt{n^4+1} + \sqrt{n^4-1}}}{\sqrt{n^4+1} + \sqrt{n^4-1}}$$

$$\therefore u_n = \frac{\left(\sqrt{n^4+1}\right)^2 - \left(\sqrt{n^4-1}\right)^2}{\left(\sqrt{n^4+1} + \sqrt{n^4-1}\right)}$$

Method:10 ⇝ Example:3(ii)(Continue)

$$\therefore u_n = \frac{\left(\sqrt{n^4+1}\right)^2 - \left(\sqrt{n^4-1}\right)^2}{\left(\sqrt{n^4+1}+\sqrt{n^4-1}\right)}$$

$$\therefore u_n = \frac{\cancel{n^4}+1-\cancel{n^4}+1}{\left(\sqrt{n^4+1}+\sqrt{n^4-1}\right)}$$

$$\therefore u_n = \frac{2}{\sqrt{n^4}\left(\sqrt{1+\frac{1}{n^4}}+\sqrt{1-\frac{1}{n^4}}\right)}$$

$$\therefore u_n = \frac{2}{n^2\left(\sqrt{1+\frac{1}{n^4}}+\sqrt{1-\frac{1}{n^4}}\right)}$$

➡ Let, $v_n = \frac{1}{n^2}$

➡ Now, $\lim\limits_{n\to\infty} \frac{u_n}{v_n}$

Complete this example by yourself.

Method:10 → Limit Comparison Test

Example:4 Check the convergence of (i). $\sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right)$.

Solution: Here, $u_n = \tan\left(\frac{1}{n}\right)$

→ Let, $v_n = \frac{1}{n}$

→ Now, $\lim_{n\to\infty} \frac{u_n}{v_n} = \lim_{n\to\infty} \frac{\tan\left(\frac{1}{n}\right)}{\frac{1}{n}}$

➡ Now, $\lim_{n\to\infty}\frac{u_n}{v_n} = \lim_{n\to\infty}\frac{\tan\left(\frac{1}{n}\right)}{\frac{1}{n}}$

$$= \lim_{\frac{1}{n}\to 0}\frac{\tan\left(\frac{1}{n}\right)}{\frac{1}{n}} \quad \left(\because \text{as } n\to\infty, \frac{1}{n}\to 0\right)$$

$= 1$ which is finite and non-zero

Therefore, by **limit comparison test** both the series are convergent or divergent together.

Method:10 ⇝ Example:4(i)(Continue)

➡ Now, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n}$ which is a p-series with $\mathbf{p = 1}$

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

➡ Thus, by **limit comparison test** the given series $\sum_{n=1}^{\infty} \tan\left(\frac{1}{n}\right)$ is divergent.

Method:10 → Limit Comparison Test

Example:4 Check the convergence of (ii). $\sum_{n=1}^{\infty} \frac{1}{n}\sin\left(\frac{1}{n}\right)$.

Solution: Here, $u_n = \frac{1}{n}\sin\left(\frac{1}{n}\right)$

→ Let, $v_n = \frac{1}{n^2}$

→ Now, $\lim\limits_{n \to \infty} \frac{u_n}{v_n} = \lim\limits_{n \to \infty} \frac{\frac{1}{n}\sin\left(\frac{1}{n}\right)}{\frac{1}{n^2}}$

Method:10 ⇝ Example:4(ii)(Continue)

Now, $\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{\frac{1}{n}\sin\left(\frac{1}{n}\right)}{\frac{1}{n^2}}$

$$=\lim_{n\to\infty}\frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}}$$

$$=\lim_{\frac{1}{n}\to 0}\frac{\sin\left(\frac{1}{n}\right)}{\frac{1}{n}}\quad\left(\because \text{as } n\to\infty,\frac{1}{n}\to 0\right)$$

$= 1$ which is finite and non-zero

Method:10 ⇝ Example:4(i)(Continue)

$= 1$ which is finite and non-zero

Therefore, by **limit comparison test** both the series are convergent or divergent together.

- Now, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ which is a p-series with $\mathbf{p = 2 > 1}$

  So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

- Thus, by **limit comparison test** the given series $\sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{1}{n}\right)$ is convergent.

Formulae…

$$1+2+3+4+5+\cdots+n=\sum_{k=1}^{n} k=\frac{n(n+1)}{2}$$

$$1^2+2^2+3^2+4^2+5^2+\cdots+n^2=\sum_{k=1}^{n} k^2=\frac{n(n+1)(2n+1)}{6}$$

$$1^3+2^3+3^3+4^3+5^3+\cdots+n^3=\sum_{k=1}^{n} k^3=\left[\frac{n(n+1)}{2}\right]^2$$

Example:5 Test the convergence of $\sum_{n=1}^{\infty} \frac{1}{1^2+2^2+3^2+\cdots+n^2}$.

Solution: Here, $\sum_{n=1}^{\infty} \frac{1}{1^2+2^2+3^2+\cdots+n^2}$.

$\sum_{k=1}^{n} k^2$

$$= \sum_{n=1}^{\infty} \frac{1}{\sum_{k=1}^{n} k^2}$$

$$= \sum_{n=1}^{\infty} \frac{1}{\sum_{k=1}^{n} k^2}$$

$$= \sum_{n=1}^{\infty} \frac{6}{n(n+1)(2n+1)}$$

Here, $u_n = \dfrac{6}{n\boxed{(n+1)}\boxed{(2n+1)}}$

$$\therefore u_n = \frac{6}{\boxed{n^3}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}$$

Let, $v_n = \dfrac{1}{n^3}$

$$\because \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$

Now, $\displaystyle \lim_{n\to\infty} \frac{6 \cdot \cancel{n^3}}{\cancel{n^3}\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)}$

$n \cdot n \cdot n =$

Method:10 ⇝ Example:5(Continue)

$$= \lim_{n \to \infty} \frac{6 \cdot \cancel{n^3}}{\cancel{n^3}\left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right)}$$

$$= \frac{6}{2} \qquad \left\{\therefore \lim_{n \to \infty}\left(\frac{1}{n}\right) = 0\right\}$$

$= 3$ which is finite and non-zero

Therefore, by **limit comparison test** both the series are convergent or divergent together.

Method:10 ⇝ Example:5(Continue)

Therefore, by **limit comparison test** both the series are convergent or divergent together.

- Now, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n^3}$ which is a p-series with $\mathbf{p = 3 > 1}$

  So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent.

- Thus, by **limit comparison test** the given series is convergent.

Method:10 → Limit Comparison Test

Example:6 Test the convergence of $\frac{1}{1^2-3}+\frac{3}{2^2-3}+\frac{5}{3^2-3}+\cdots$.

Solution: Here, $\frac{1}{1^2-3}+\frac{3}{2^2-3}+\frac{5}{3^2-3}+\cdots=\sum_{n=1}^{\infty}\frac{2n-1}{n^2-3}$

Here, $u_n=\frac{2n-1}{n^2-3}$

$$\therefore u_n=\frac{n\left(2-\frac{1}{n}\right)}{n^2\left(1-\frac{3}{n^2}\right)}$$

Method:10 ⇝ Example:6(Continue)

$$\therefore u_n = \frac{n\left(2-\frac{1}{n}\right)}{n^2\left(1-\frac{3}{n^2}\right)}$$

$$\therefore u_n = \frac{\left(2-\frac{1}{n}\right)}{n\left(1-\frac{3}{n^2}\right)}$$

➡ $\text{Let}, v_n = \frac{1}{n}$

Complete this example by yourself.

Method:11 → D'Alembert's Ratio Test

D'Alembert's Ratio Test

- Let $\sum u_n$ be the positive term series and $\lim\limits_{n \to \infty} \left| \frac{u_n}{u_{n+1}} \right| = L$ and

1) If $L > 1$, then the given series is **convergent**.
2) If $0 \leq L < 1$, then the given series is **divergent**.
3) If $L = 1$, then the **ratio test fails**.

D'Alembert's Ratio Test

$$\sum_{n=a}^{\infty} u_n$$

$$\lim_{n \to \infty} \left| \frac{u_n}{u_{n+1}} \right| = L$$

- $L > 1$ then **convergent**
- $L = 1$ then **the test fails**
- $0 \leq L < 1$ then **divergent**

Method:11 → Ratio Test

Example:1 Check the convergence of (1) $\sum_{n=1}^{\infty} \frac{2^n}{n^2}$.

Solution: Here, $u_n = \frac{2^n}{n^2} \Rightarrow u_{n+1} = \frac{2^{n+1}}{(n+1)^2}$

→ Now, $\lim_{n \to \infty} \frac{u_n}{u_{n+1}} = \lim_{n \to \infty} \frac{2^n \cdot (n+1)^2}{n^2 \cdot 2^{n+1}}$

$$= \lim_{n \to \infty} \frac{\cancel{2^n} \cdot \cancel{n^2} \left(1 + \frac{1}{n}\right)^2}{\cancel{n^2} \cdot \cancel{2^n} \cdot 2}$$

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Now, $\lim\limits_{n \to \infty} \dfrac{u_n}{u_{n+1}} = \lim\limits_{n \to \infty} \dfrac{2^n \cdot (n+1)^2}{n^2 \cdot 2^{n+1}}$

$$= \lim_{n \to \infty} \frac{\cancel{2^n} \cdot \cancel{n^2}\left(1 + \frac{1}{n}\right)^2}{\cancel{n^2} \cdot \cancel{2^n} \cdot 2}$$

$$= \frac{1}{2} < \mathbf{1}$$

Thus, by **ratio test**, $\sum\limits_{n=1}^{\infty} \dfrac{2^n}{n^2}$ is divergent.

Method:11 → Ratio Test

Example:1 Check the convergence of (2) $\sum_{n=1}^{\infty} \frac{2^n - 1}{3^n}$.

Solution: Here, $u_n = \frac{2^n - 1}{3^n} \Rightarrow u_{n+1} = \frac{2^{n+1} - 1}{3^{n+1}}$

→ Now, $\lim_{n \to \infty} \frac{u_n}{u_{n+1}} = \lim_{n \to \infty} \frac{(2^n - 1) \cdot 3^{n+1}}{3^n \cdot (2^{n+1} - 1)}$

$$= \lim_{n \to \infty} \frac{(2^n - 1) \cdot 3^n \cdot 3}{3^n \cdot (2 \cdot 2^n - 1)}$$

Method:11 ⇝ Example:1(2)(Continue)

$$= \lim_{n \to \infty} \frac{(2^n - 1) \cdot \cancel{3^n} \cdot 3}{\cancel{3^n} \cdot (2 \cdot 2^n - 1)}$$

$$= \lim_{n \to \infty} \frac{3 \cdot \cancel{2^n} \left(1 - \frac{1}{2^n}\right)}{\cancel{2^n} \cdot \left(2 - \frac{1}{2^n}\right)}$$

$$= \lim_{n \to \infty} \frac{3 \cdot \left(1 - \frac{1}{2^n}\right)}{\left(2 - \frac{1}{2^n}\right)}$$

$$= \frac{3(1 - 0)}{2 - 0}$$

$$= \frac{3}{2} > \mathbf{1}$$

⇒ Thus, by **ratio test**, $\sum_{n=1}^{\infty} \frac{2^n - 1}{3^n}$ is convergent.

Method:11 → Ratio Test

Example:2 State D'Alembert's ratio test. Discuss the convergence of

$$\sum_{n=1}^{\infty} \frac{4^n (n+1)!}{n^{n+1}}.$$

Solution: Here, $u_n = \dfrac{4^n (n+1)!}{n^{n+1}} \Rightarrow u_{n+1} = \dfrac{4^{n+1} (n+2)!}{(n+1)^{n+2}}$

Now, $\displaystyle \lim_{n \to \infty} \frac{u_n}{u_{n+1}} = \lim_{n \to \infty} \frac{4^n (n+1)! \cdot (n+1)^{n+2}}{n^{n+1} \cdot 4^{n+1} (n+2)!}$

Method:11 ⇝ Example:2(Continue)

5! = 5 · 4 · 3 · 2 · 1

$= 5 \cdot (4 \cdot 3 \cdot 2 \cdot 1)$

$= 5 \cdot 4!$

$(n+2)! = (n+2) \cdot (n+1) \cdot (n) \cdots 2 \cdot 1$

$= (n+2) \cdot (n+1)!$

➡ Now, $\displaystyle \lim_{n \to \infty} \frac{u_n}{u_{n+1}} = \lim_{n \to \infty} \frac{4^n (n+1)! \cdot \boxed{(n+1)^{n+2}}}{\boxed{n^{n+1}} \cdot \boxed{4^{n+1}} \boxed{(n+2)!}}$

$$= \lim_{n \to \infty} \frac{\cancel{4^n}\, \cancel{(n+1)!}\, (n+1)^n\, (n+1)^2}{n^n \cdot n \cdot \cancel{4^n} \cdot 4\, (n+2)\, \cancel{(n+1)!}}$$

$$= \lim_{n \to \infty} \frac{(n+1)^n (n+1)^2}{n^n \cdot 4n \cdot (n+2)}$$

$$= \lim_{n \to \infty} \frac{\cancel{n^n}\left(1+\frac{1}{n}\right)^n \cancel{n^2}\left(1+\frac{1}{n}\right)^2}{\cancel{n^n} \cdot 4\cancel{n} \cdot \cancel{n}\left(1+\frac{2}{n}\right)}$$

Method:11 ⇝ Example:2(Continue)

$$= \lim_{n\to\infty} \frac{\cancel{n^n}\left(1+\frac{1}{n}\right)^n \cancel{n^2}\left(1+\frac{1}{n}\right)^2}{\cancel{n^n}\cdot 4\cancel{n}\cdot\cancel{n}\left(1+\frac{2}{n}\right)}$$

$$= \lim_{n\to\infty} \frac{\left(1+\frac{1}{n}\right)^n \left(1+\frac{1}{n}\right)^2}{4\left(1+\frac{2}{n}\right)}$$

➡ We know that, $\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x$

$$= \frac{e\,(1+0)^2}{4(1+0)}$$

$$= \frac{e}{4} < 1 \quad \{\because e \approx 2.71\}$$

➡ Thus, by **ratio test**,

$\sum_{n=1}^{\infty} \frac{4^n (n+1)!}{n^{n+1}}$ is divergent.

Method:11 Ratio Test

Example:3 Check the convergence of $1+\frac{2^p}{2!}+\frac{3^p}{3!}+\frac{4^p}{4!}+\cdots$

Solution: Here, $1+\frac{2^p}{2!}+\frac{3^p}{3!}+\frac{4^p}{4!}+\cdots=\sum_{n=1}^{\infty}\frac{n^p}{n!}$

- Here, $u_n=\frac{n^p}{n!} \Rightarrow u_{n+1}=\frac{(n+1)^p}{(n+1)!}$
- Now, $\lim_{n\to\infty}\frac{u_n}{u_{n+1}}=\lim_{n\to\infty}\frac{n^p\cdot(n+1)!}{n!\cdot(n+1)^p}$

Method:11 ⇝ Example:3(Continue)

➡ Now, $\lim\limits_{n\to\infty}\frac{u_n}{u_{n+1}} = \lim\limits_{n\to\infty}\frac{n^p\cdot(n+1)!}{n!\cdot(n+1)^p}$

$$= \lim_{n\to\infty}\frac{n^p\cdot(n+1)\cdot n!}{n!\cdot n^p\left(1+\frac{1}{n}\right)^p}$$

$$= \lim_{n\to\infty}\frac{(n+1)}{\left(1+\frac{1}{n}\right)^p}$$

$$= \lim_{n\to\infty}\frac{n\cdot\left(1+\frac{1}{n}\right)}{\left(1+\frac{1}{n}\right)^p}$$

$$= \infty > 1$$

➡ Thus, by **ratio test**, $1+\frac{2^p}{2!}+\frac{3^p}{3!}+\frac{4^p}{4!}+\cdots$ is convergent.

Example:4 Check the convergence of $\frac{1}{3}+\frac{1\cdot 2}{3\cdot 5}+\frac{1\cdot 2\cdot 3}{3\cdot 5\cdot 7}+\cdots$

Solution: Here, $\frac{1}{3}+\frac{1\cdot 2}{3\cdot 5}+\frac{1\cdot 2\cdot 3}{3\cdot 5\cdot 7}+\cdots=\sum_{n=1}^{\infty}\frac{1\cdot 2\cdot 3\cdot\cdots\cdot n}{3\cdot 5\cdot 7\cdot\cdots\cdot(2n+1)}$

Here, $u_n=\frac{1\cdot 2\cdot 3\cdot\cdots\cdot n}{3\cdot 5\cdot 7\cdot\cdots\cdot(2n+1)}$

$$\Longrightarrow u_{n+1}=\frac{1\cdot 2\cdot 3\ldots n\cdot(n+1)}{3\cdot 5\cdot 7\ldots(2n+1)(2n+3)}$$

Complete this example by yourself.

Example:5 Check the convergence of (1) $\sum_{n=1}^{\infty} \frac{3^n + 4^n}{4^n + 5^n}$.

Solution: Here, $u_n = \frac{3^n + 4^n}{4^n + 5^n} \Longrightarrow u_{n+1} = \frac{3^{n+1} + 4^{n+1}}{4^{n+1} + 5^{n+1}}$

➡ Now, $\lim_{n \to \infty} \frac{u_n}{u_{n+1}} = \lim_{n \to \infty} \frac{\boxed{(3^n + 4^n)} \cdot \boxed{(4^{n+1} + 5^{n+1})}}{\boxed{(3^{n+1} + 4^{n+1})} \cdot \boxed{(4^n + 5^n)}}$

$$= \lim_{n \to \infty} \frac{4^n \left(\frac{3^n}{4^n} + 1\right) \cdot 5^{n+1} \left(\frac{4^{n+1}}{5^{n+1}} + 1\right)}{4^{n+1} \left(\frac{3^{n+1}}{4^{n+1}} + 1\right) \cdot 5^n \left(\frac{4^n}{5^n} + 1\right)}$$

Method:11 ⇝ Example:5(Continue)

$$= \lim_{n \to \infty} \frac{4^n\left(\frac{3^n}{4^n}+1\right)\cdot 5^{n+1}\left(\frac{4^{n+1}}{5^{n+1}}+1\right)}{4^{n+1}\left(\frac{3^{n+1}}{4^{n+1}}+1\right)\cdot 5^n\left(\frac{4^n}{5^n}+1\right)}$$

$$= \lim_{n \to \infty} \frac{\cancel{4^n}\left(\frac{3^n}{4^n}+1\right)\cdot 5\cdot \cancel{5^n}\left(\frac{4^{n+1}}{5^{n+1}}+1\right)}{4\cdot \cancel{4^n}\left(\frac{3^{n+1}}{4^{n+1}}+1\right)\cdot \cancel{5^n}\left(\frac{4^n}{5^n}+1\right)}$$

$$= \lim_{n \to \infty} \frac{\cancel{4^n}\left(\frac{3^n}{4^n}+1\right)\cdot 5\cdot \cancel{5^n}\left(\frac{4^{n+1}}{5^{n+1}}+1\right)}{4\cdot \cancel{4^n}\left(\frac{3^{n+1}}{4^{n+1}}+1\right)\cdot \cancel{5^n}\left(\frac{4^n}{5^n}+1\right)}$$

$$= \lim_{n \to \infty} \frac{5\cdot\left(\left(\frac{3}{4}\right)^{n}+1\right)\cdot\left(\left(\frac{4}{5}\right)^{n+1}+1\right)}{4\cdot\left(\left(\frac{3}{4}\right)^{n+1}+1\right)\cdot\left(\left(\frac{4}{5}\right)^{n}+1\right)}$$

$$= \lim_{n \to \infty} \frac{5 \cdot \left(\left(\frac{3}{4}\right)^{n} + 1\right) \cdot \left(\left(\frac{4}{5}\right)^{n+1} + 1\right)}{4 \cdot \left(\left(\frac{3}{4}\right)^{n+1} + 1\right) \cdot \left(\left(\frac{4}{5}\right)^{n} + 1\right)}$$

We know that, $\lim_{n \to \infty} a^n = 0$; if $-1 < a < 1$

$$= \frac{5 \cdot (0 + 1) \cdot (0 + 1)}{4 \cdot (0 + 1) \cdot (0 + 1)}$$

$$= \frac{5}{4} > 1$$

Thus, by **ratio test**,

$\sum_{n=1}^{\infty} \frac{3^n + 4^n}{4^n + 5^n}$ is convergent.

Method:11 → Ratio Test

Example:6 Test the convergence of the series $\dfrac{1}{1\cdot 2\cdot 3}+\dfrac{x}{4\cdot 5\cdot 6}+\dfrac{x^2}{7\cdot 8\cdot 9}+\dfrac{x^3}{10\cdot 11\cdot 12}\ldots,\ x\geq 0.$

Solution: Here, $\dfrac{1}{1\cdot 2\cdot 3}+\dfrac{x}{4\cdot 5\cdot 6}+\dfrac{x^2}{7\cdot 8\cdot 9}+\dfrac{x^3}{10\cdot 11\cdot 12}+\cdots$

$$=\sum_{n=0}^{\infty}\frac{x^n}{(3n+1)(3n+2)(3n+3)}$$

Method:11 ⇝ Example:6(Continue)

$$= \sum_{n=0}^{\infty} \frac{x^n}{(3n+1)(3n+2)(3n+3)}$$

Here, $u_n = \dfrac{x^n}{(3n+1)(3n+2)(3n+3)} \Longrightarrow u_{n+1} = \dfrac{x^{n+1}}{(3n+4)(3n+5)(3n+6)}$

Now, $\displaystyle\lim_{n\to\infty} \frac{u_n}{u_{n+1}} = \lim_{n\to\infty} \frac{x^n \cdot (3n+4)(3n+5)(3n+6)}{x^{n+1} \cdot (3n+1)(3n+2)(3n+3)}$

$$= \lim_{n\to\infty} \frac{(3n+4)(3n+5)(3n+6)}{x \cdot (3n+1)(3n+2)(3n+3)}$$

$$= \lim_{n \to \infty} \frac{(3n+4)(3n+5)(3n+6)}{x \cdot (3n+1)(3n+2)(3n+3)}$$

$$= \lim_{n \to \infty} \frac{n \cdot \left(3 + \frac{4}{n}\right) \cdot n \cdot \left(3 + \frac{5}{n}\right) \cdot n \cdot \left(3 + \frac{6}{n}\right)}{x \cdot n \cdot \left(3 + \frac{1}{n}\right) \cdot n \cdot \left(3 + \frac{2}{n}\right) \cdot n \cdot \left(3 + \frac{3}{n}\right)}$$

$$= \frac{(3+0)(3+0)(3+0)}{x(3+0)(3+0)(3+0)}$$

$$= \frac{1}{x} = L$$

Method:11 ⇝ Example:6(Continue)

$$= \frac{1}{x} = L$$

Since, $x \geq 0$

Case:1 If $\mathbf{x > 1} \Longrightarrow \frac{1}{x} = L < 1$

Thus, the given series is **divergent** by D'Alembert's ratio test.

Case:2 If $\mathbf{0 < x < 1} \Longrightarrow \frac{1}{x} = L > 1$

Thus, the given series is **convergent** by D'Alembert's ratio test.

Case:3 If $x = 1 \Longrightarrow \frac{1}{x} = L = 1$

D'Alembert's ratio test fails.

For $x = 1$

$$\sum_{n=0}^{\infty} \frac{1}{(3n+1)(3n+2)(3n+3)}$$

$$\sum_{n=0}^{\infty} \frac{x^n}{(3n+1)\,(3n+2)\,(3n+3)}$$

Here, $u_n = \dfrac{1}{(3n+1)(3n+2)(3n+3)}$

Method:11 ⇝ Example:6(Continue)

Here, $u_n = \dfrac{1}{(3n+1)(3n+2)(3n+3)}$

$$\Rightarrow u_n = \frac{1}{n^3\left(3+\frac{1}{n}\right)\left(3+\frac{2}{n}\right)\left(3+\frac{3}{n}\right)}$$

- Let, $v_n = \dfrac{1}{n^3}$

- Now, $\lim\limits_{n \to \infty} \dfrac{u_n}{v_n}$

➡ Now, $\lim_{n \to \infty} \frac{u_n}{v_n}$

$$= \lim_{n \to \infty} \frac{\cancel{n^3}}{\cancel{n^3}\left(3 + \frac{1}{n}\right)\left(3 + \frac{2}{n}\right)\left(3 + \frac{3}{n}\right)}$$

$$= \frac{1}{(3+0)(3+0)(3+0)}$$

$= \frac{1}{27}$ which is finite and non-zero

$= \dfrac{1}{27}$ which is finite and non-zero

Therefore, by **limit comparison test** both the series are convergent or divergent together.

➡ Now, $\sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \dfrac{1}{n^3}$ which is a p-series with $\mathbf{p = 3 > 1}$

So, by **p-series** test, $\sum_{n=1}^{\infty} \dfrac{1}{n^3}$ is convergent.

So, by **p-series** test, $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent.

➡ Thus, by **limit comparison test** series $\sum_{n=0}^{\infty} \frac{1}{(3n+1)(3n+2)(3n+3)}$

is convergent when $x = 1$.

Conclusion The given series is **convergent** for $x \geq 1$ and **divergent** for $x < 1$.

Method:13 ⟶ Cauchy's n^{th} Root(Radical) Test

Cauchy's n^{th} Root Test

$\lim\limits_{n \to \infty} \sqrt[n]{u_n}$

- Let $\sum u_n$ be the positive term series and $\lim\limits_{n \to \infty} (u_n)^{\frac{1}{n}} = L$ and

1) If $0 \le L < 1$, then the given series is **convergent**.

2) If $L > 1$, then the given series is **divergent**.

3) If $L = 1$, then the **root test fails**.

Method:13 → Cauchy's n^{th} Root Test

Example:1 Check the convergence of $\sum_{n=1}^{\infty} \frac{(n+\sqrt{n})^n}{3^n \cdot n^{n+1}}$.

Solution: Here, $u_n = \frac{(n+\sqrt{n})^n}{3^n \cdot n^{n+1}}$

→ Now, $\lim_{n \to \infty} (u_n)^{\frac{1}{n}} = \lim_{n \to \infty} \left(\frac{(n+\sqrt{n})^n}{3^n \cdot n^{n+1}} \right)^{\frac{1}{n}}$

$$= \lim_{n \to \infty} \left(\frac{(n+\sqrt{n})^n}{3^n \cdot n \cdot n^n} \right)^{\frac{1}{n}}$$

Method:13 ⇝ Example:1(Continue)

$$= \lim_{n \to \infty} \left(\frac{(n + \sqrt{n})^n}{3^n \cdot n \cdot n^n} \right)^{\frac{1}{n}}$$

$$= \lim_{n \to \infty} \frac{(n + \sqrt{n})^{n \cdot \frac{1}{n}}}{(3^n)^{\frac{1}{n}} \cdot (n^n)^{\frac{1}{n}} \cdot n^{\frac{1}{n}}}$$

$$= \lim_{n \to \infty} \frac{(n + \sqrt{n})}{3n \cdot n^{\frac{1}{n}}}$$

$$= \lim_{n \to \infty} \frac{n\left(1 + \frac{\sqrt{n}}{n}\right)}{3n \cdot n^{\frac{1}{n}}}$$

$$= \lim_{n \to \infty} \frac{\left(1 + \frac{1}{\sqrt{n}}\right)}{3 \cdot n^{\frac{1}{n}}}$$

$$= \frac{1}{3 \cdot \lim_{n \to \infty} n^{\frac{1}{n}}}$$

Method:13 ⇝ Example:1(Continue)

$$= \frac{1}{3 \cdot \lim\limits_{n \to \infty} n^{\frac{1}{n}}}$$

$$= \frac{1}{3} \cdot \lim_{n \to \infty} n^{-\frac{1}{n}} \dashrightarrow \textbf{(1)}$$

Where, $\lim\limits_{n \to \infty} n^{-\frac{1}{n}} \longrightarrow (\infty^0)$

$$\text{Assume } y = \lim_{n \to \infty} (n)^{\left(-\frac{1}{n}\right)}$$

$$\log y = \log \left\{ \lim_{n \to 0} (n)^{\left(-\frac{1}{n}\right)} \right\}$$

$$= \lim_{n \to \infty} \left\{ \log(n)^{\left(-\frac{1}{n}\right)} \right\}$$

$$= \lim_{n \to \infty} \left\{ \left(-\frac{1}{n} \right) \cdot \log(n) \right\} (0 \times \infty)$$

$$= \lim_{n \to \infty} -\left\{ \frac{\log(n)}{n} \right\} \left(\frac{0}{0} \right)$$

$$= \lim_{n \to \infty} -\left\{\frac{\log(n)}{n}\right\} \left(\frac{0}{0}\right)$$

Applying L'Hospital's Rule…

$$= \lim_{n \to \infty} -\left\{\frac{\frac{1}{n}}{1}\right\}$$

$$= 0$$

$$\because \lim_{n \to \infty} \left(\frac{1}{n}\right) = 0$$

$$\therefore \log y = 0$$

$$\Longrightarrow y = e^0$$

$$\Longrightarrow y = 1$$

$$\therefore \lim_{n \to \infty} (n)^{\left(-\frac{1}{n}\right)} = 1$$

From equation **(1)**

$$\lim_{n \to \infty} (u_n)^{\frac{1}{n}} = \frac{1}{3} \cdot \lim_{n \to \infty} n^{-\frac{1}{n}}$$

$$= \frac{1}{3} < 1$$

Thus, by **Cauchy's n^{th} root test**, the given series is convergent.

Method:13 → Cauchy's n^{th} Root Test

Example:2 Check the convergence of

$$\left(\frac{2^2}{1^2}-\frac{2^1}{1^1}\right)^{-1}+\left(\frac{3^3}{2^3}-\frac{3^1}{2^1}\right)^{-2}+\left(\frac{4^4}{3^4}-\frac{4^1}{3^1}\right)^{-3}+\cdots$$

Solution: Here, $\left(\frac{2^2}{1^2}-\frac{2^1}{1^1}\right)^{-1}+\left(\frac{3^3}{2^3}-\frac{3^1}{2^1}\right)^{-2}+\left(\frac{4^4}{3^4}-\frac{4^1}{3^1}\right)^{-3}+\cdots$

$$=\sum_{n=1}^{\infty}\left(\frac{(n+1)^{n+1}}{n^{n+1}}-\frac{(n+1)^1}{n^1}\right)^{-n}$$

Method:13 ⇝ Example:2(Continue)

$$= \sum_{n=1}^{\infty} \left(\frac{(n+1)^{n+1}}{n^{n+1}} - \frac{(n+1)^{1}}{n^{1}} \right)^{-n}$$

Here, $u_n = \left(\frac{(n+1)^{n+1}}{n^{n+1}} - \frac{(n+1)^{1}}{n^{1}} \right)^{-n}$

$$\Rightarrow u_n = \left[\left(\frac{n+1}{n} \right)^{n+1} - \left(\frac{n+1}{n} \right) \right]^{-n}$$

Method:13 ⇝ Example:2(Continue)

$$\Rightarrow u_n = \left[\left(\frac{n+1}{n}\right)^{n+1} - \left(\frac{n+1}{n}\right)\right]^{-n}$$

$$\Rightarrow (u_n)^{\frac{1}{n}} = \left[\left(\frac{n+1}{n}\right)^{n+1} - \left(\frac{n+1}{n}\right)\right]^{-1}$$

➡ Now, $\lim\limits_{n\to\infty} (u_n)^{\frac{1}{n}} = \lim\limits_{n\to\infty} \left[\left(\frac{n+1}{n}\right)^{n+1} - \left(\frac{n+1}{n}\right)\right]^{-1}$

Method:13 ⇝ Example:2(Continue)

➡ Now, $\lim_{n \to \infty} (u_n)^{\frac{1}{n}} = \lim_{n \to \infty} \left[\left(\frac{n+1}{n} \right)^{n+1} - \left(\frac{n+1}{n} \right) \right]^{-1}$

$$= \lim_{n \to \infty} \left[\left(\frac{n+1}{n} \right)^{n} \cdot \left(\frac{n+1}{n} \right) - \left(\frac{n+1}{n} \right) \right]^{-1}$$

$$= \lim_{n \to \infty} \left[\left(1 + \frac{1}{n} \right)^{n} \cdot \left(1 + \frac{1}{n} \right) - \left(1 + \frac{1}{n} \right) \right]^{-1}$$

$$= \left[\lim_{n \to \infty} \left(1 + \frac{1}{n} \right)^{n} \cdot \lim_{n \to \infty} \left(1 + \frac{1}{n} \right) - \lim_{n \to \infty} \left(1 + \frac{1}{n} \right) \right]^{-1}$$

$$= \left[\boxed{\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n}} \cdot \lim_{n\to\infty}\left(1+\frac{1}{n}\right) - \lim_{n\to\infty}\left(1+\frac{1}{n}\right)\right]^{-1}$$

$$\left\{\because \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^{n} = e^{x}\right\}$$

$$= [e^{1} \cdot (1+0) - (1+0)]^{-1}$$

$$= [e-1]^{-1}$$

$$= \frac{1}{e-1} < \mathbf{1}$$

➡ Thus, by **Cauchy's n^{th} root test**, the given series is **convergent**.

Example:3 Check the convergence of $1+\left(\frac{2}{3}\right)x+\left(\frac{3}{4}\right)^2x^2+\left(\frac{4}{5}\right)^3x^3+\cdots \; ; x>0$

Solution: Here, $1+\left(\frac{2}{3}\right)x+\left(\frac{3}{4}\right)^2x^2+\left(\frac{4}{5}\right)^3x^3+\cdots$

$$=\sum_{n=0}^{\infty}\left(\frac{n+1}{n+2}\right)^n\cdot x^n$$

Here, $u_n=\left(\frac{n+1}{n+2}\right)^n\cdot x^n$

Here, $u_n = \left(\frac{n+1}{n+2}\right)^n \cdot x^n$

$$\Rightarrow (u_n)^{\frac{1}{n}} = \left[\left(\frac{n+1}{n+2}\right)^n \cdot x^n\right]^{\frac{1}{n}}$$

$$= \left[\left(\frac{n+1}{n+2}\right)^{\frac{n}{n}} \cdot x^{\frac{n}{n}}\right]$$

$$= \left[\left(\frac{n+1}{n+2}\right) \cdot x\right]$$

Now, $\lim_{n \to \infty} (u_n)^{\frac{1}{n}}$

$$= \lim_{n \to \infty} \left[\left(\frac{n+1}{n+2}\right) \cdot x\right]$$

$$= \lim_{n \to \infty} \left[\left(\frac{1+\frac{1}{n}}{1+\frac{2}{n}}\right) \cdot x\right]$$

$$= \left[\left(\frac{1+0}{1+0}\right) \cdot x\right]$$

$$= \left[\left(\frac{1+0}{1+0}\right)\cdot x\right] = x = L$$

$$\because \lim_{n\to\infty}\left(\frac{1}{n}\right) = 0$$

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➡ Thus, by **Cauchy's n^{th} root test**, the given series is **convergent** if $L = x < 1$ and is **divergent** if $L = x > 1$.

But the Root test fail for $x = 1$

➡ For $x = 1$, $\sum_{n=0}^{\infty}\left(\frac{n+1}{n+2}\right)^n$

$$\therefore u_n = \left(\frac{n+1}{n+2}\right)^n$$

Cauchy's n^{th} Root Test

▷ Let $\sum u_n$ be the positive term series and $\lim_{n\to\infty}(u_n)^{\frac{1}{n}} = L$ and

1) If $0 \le L < 1$, then the given series is **convergent**.
2) If $L > 1$, then the given series is **divergent**.
3) If $L = 1$, then the **root test fails**.

$$\therefore u_n = \left(\frac{n+1}{n+2}\right)^n$$

$$\Rightarrow u_n = \left(\frac{1+\frac{1}{n}}{1+\frac{2}{n}}\right)^n$$

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty}\left(\frac{1+\frac{1}{n}}{1+\frac{2}{n}}\right)^n$$

$$= \frac{\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n}{\lim_{n\to\infty}\left(1+\frac{2}{n}\right)^n}$$

$$\left\{\therefore \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x\right\}$$

$$= \frac{e}{e^2}$$

$$= \frac{1}{e} \neq 0$$

Method:13 ⇝ Example:3(Continue)

$$= \frac{1}{e} \neq 0$$

➡ Thus, by **zero test**, the given series is **divergent** if $x = 1$.

Conclusion

The given series is **convergent** for $x < 1$ and **divergent** for $x \geq 1$.

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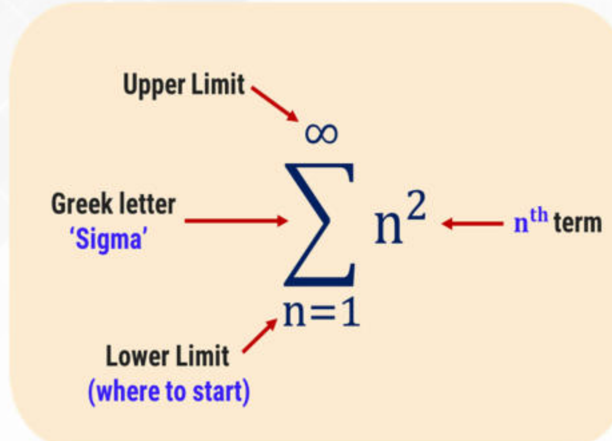

Thank You

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